

Cascade Fabric and the No-Privileged-Level Corollary: Ternary Microfluidic Networks for Partition-Depth Computation

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Abstract. We derive the ternary microfluidic cascade fabric as the unique physical implementation of multi-level categorical discrimination in a bounded aqueous partition-based computing system. The derivation proceeds from the Bounded Phase Space axiom through the No-Privileged-Level Corollary to a self-similar routing architecture of branching factor 3 and depth $k = \lceil \log_3 N_{\text{cat}} \rceil$. The No-Privileged-Level Corollary establishes that in a hierarchical partition system, no tier of the partition is intrinsically more fundamental than any other: the Koopman operator restricted to tier- n modes is unitarily equivalent to the operator restricted to tier- $(n+1)$ modes after appropriate rescaling of the phase space measure. This self-similarity forces the cascade architecture to replicate the same routing structure at every depth level, yielding a self-similar tree. The Ternary Optimality Theorem proves that the branching factor $b = 3$ minimises the total number of cascade junctions (fabrication complexity) for a three-dimensional S-entropy address space while maintaining one-to-one correspondence between S-entropy ternary digits and routing decisions. The cascade depth $k = \lceil \log_3 N_{\text{cat}} \rceil$ is shown to be necessary and sufficient (Depth Necessity Theorem): fewer levels cannot distinguish all N_{cat} categories, and more levels provide no additional discriminability beyond the S-entropy resolution. Backward trajectory completion provides the routing algorithm: each cascade level reads one ternary digit of the input's S-entropy address and routes to the matching sub-trie, completing the full classification in $O(\log_3 N_{\text{cat}})$ steps. The microfluidic implementation requires channel widths of 1–100 μm and operates in the laminar flow regime (Reynolds number $Re \ll 1$) with Péclet numbers $Pe = vL/D \sim 10$ –100, ensuring advection-dominated transport at each junction. Validation on a three-level prototype (27-category system, $k = 3$) demonstrates $94 \pm 3\%$ correct routing over 500 trial inputs, consistent with the theoretical prediction of $\kappa_{\text{arr}}^k = (1 - (1 - \bar{\kappa})^{n_c})^k$ where $\bar{\kappa} \approx 0.85$ per channel and $n_c = 44$ channels per junction.

Keywords: cascade fabric, No-Privileged-Level Corollary, ternary routing, S-entropy address, partition depth, microfluidic network, backward trajectory completion, self-similar architecture, laminar flow

1. Introduction

The state encoder (companion paper [Sachikonye, 2026a]) maps any input system $\mathcal{X}$ to its S-entropy triple $\mathbf{S}(\mathcal{X}) = (S_k, S_t, S_e) \in [0, 1]^3$. The aperture array [Sachikonye, 2026b] filters inputs by their S-entropy coordinates, passing those within distance σ of the target with probability κ . But categorical classification requires more than a single filter:

distinguishing N_{cat} categories demands a routing mechanism that directs each input to the correct sub-population among N_{cat} outputs.

The cascade fabric provides this routing. It is a hierarchical network of channel junctions, each operating an aperture-array filter on one ternary digit of the input's S-entropy address and routing to one of three outgoing branches. After k levels, inputs have been routed to one of 3^k terminal

categories.

The architecture is not chosen for engineering convenience. We prove it is *forced* by two independent arguments. First, the No-Privileged-Level Corollary (§3) establishes that the partition structure of bounded phase space is self-similar across all tiers, forcing a self-similar (tree-like) cascade architecture. Second, the Ternary Optimality Theorem (§4) proves that branching factor $b = 3$ is optimal for three-dimensional S-entropy space. Together, these force a ternary self-similar tree of depth $k = \lceil \log_3 N_{\text{cat}} \rceil$.

The microfluidic implementation (§7) follows from the aqueous medium requirement: the bilayer substrate operates in water, so the cascade must also be aqueous. In water at physiological conditions, the Reynolds number for channel widths $w \leq 100 \mu\text{m}$ is $Re \ll 1$, placing the system firmly in the laminar flow regime where routing is deterministic and controllable.

2. Bounded Phase Space Framework

Axiom 1 (Bounded Phase Space). *Every persistent physical system $\mathcal{X}$ occupies $\Omega \subset \mathbb{R}^{2d}$ with $\mu(\Omega) < \infty$, admitting a refining hierarchy of measurable partitions $\mathcal{P}_0 \supset \mathcal{P}_1 \supset \dots$ with $|\mathcal{P}_k| \rightarrow \infty$ as $k \rightarrow \infty$.*

From [Sachikonye, 2026c]: Partition coordinates (n, ℓ, m, s) enumerate the spectral states; capacity $C(n) = 2n^2$; partition depth $\mathcal{M} = \log_2 C(n)$.

From [Sachikonye, 2026a]: S-entropy coordinates $\mathbf{S} = (S_k, S_t, S_e) \in [0, 1]^3$ are the unique minimal sufficient statistics; the S-entropy address is the base-3 expansion $\mathbf{S} = \sum_{j=1}^{\infty} a_j 3^{-j}$ where $a_j \in \{0, 1, 2\}$.

From [Sachikonye, 2026b]: Aperture catalytic power $\kappa \in [0, 1]$ is a Gaussian function of S-entropy distance; parallel array saturation: $\kappa_{\text{arr}} = 1 - \prod_i (1 - \kappa_i) \rightarrow 1$ iff $\sum_i \kappa_i = \infty$.

From [Sachikonye, 2026d]: Floor Theorem: $\Delta E_{\text{op}} \geq k_B T \ln 2$; dx/dATP Identity: each level costs $\leq 1/18.2$ ATP at 100% efficiency.

3. The No-Privileged-Level Corollary

3.1. Statement

Definition 3.1 (Tier- n Restriction). *The tier- n restriction of the Koopman operator $\mathcal{U}^t$ on $L^2(\Omega, \mu)$ is its compression to the subspace spanned by the Koopman eigenfunctions at tier n : $\mathcal{U}_n^t = \Pi_n \mathcal{U}^t \Pi_n$,*

where Π_n is the spectral projection onto modes with principal partition number equal to n .

Theorem 3.2 (No-Privileged-Level Corollary). *For any two tiers $n, n' \in \mathbb{N}$, the tier- n and tier- n' Koopman operators $\mathcal{U}_n^t$ and $\mathcal{U}_{n'}^t$ are unitarily equivalent after rescaling the phase space measure:*

$$\mathcal{U}_n^t \sim_U \mathcal{U}_{n'}^t \quad \text{mod } \mu \rightarrow \mu_{n,n'}, \quad (1)$$

where $\mu_{n,n'}$ is the measure rescaled by the ratio $C(n)/C(n') = n^2/n'^2$. No tier is dynamically more fundamental than any other.

Proof. The capacity $C(n) = 2n^2$ counts the states at tier n . The Koopman eigenfunctions at tier n form an orthonormal basis for the $C(n)$ -dimensional eigenspace of $\mathcal{U}_n^t$. Define the unitary intertwining operator $W_{n,n'} : \text{ran}(\Pi_n) \rightarrow \text{ran}(\Pi_{n'})$ by mapping the j -th basis vector at tier n to the $\lfloor j \cdot C(n')/C(n) \rfloor$ -th basis vector at tier n' (reindexed to match dimensions via the measure rescaling $\mu \rightarrow \mu_{n,n'}$ that normalises both spaces to unit volume).

Since the Koopman operator is characterised entirely by its eigenvalues $\{e^{i\omega_k t}\}$ and since the eigenvalue structure at tier n maps bijectively to the eigenvalue structure at tier n' under the frequency rescaling $\omega_k^{(n)} \rightarrow (n/n')^2 \omega_k^{(n)}$ (which preserves the discrete spectrum structure and maps the tier- n partition cell structure to the tier- n' structure), the operators are unitarily equivalent up to this rescaling.

In particular, the partition operations (transitions between cells within a tier) at tier n are structurally identical to those at tier n' : the routing decision at depth n of the cascade is the same type of operation as at depth n' , with the same algebraic structure and the same Landauer cost $k_B T \ln 3$ per ternary step (Theorem 6.1 of [Sachikonye, 2026d]). $\square$

Corollary 3.3 (Architectural Self-Similarity). *The cascade fabric implementing partition-based routing must have identical structure at every depth level. The routing junction at depth j is a copy of the routing junction at depth $j + 1$, scaled by the measure ratio $C(n_j)/C(n_{j+1})$.*

Proof. If the routing operation at depth j differed structurally from that at depth $j + 1$, then tier n_j would be a privileged tier (one requiring a different type of computation). Theorem 3.2 excludes this: no tier is privileged, so the routing junction must

have the same algebraic structure at every depth. The only constraint is the measure scaling, which in the microfluidic implementation corresponds to a channel-width scaling (wider channels for earlier tiers, narrower for later tiers). $\square$

Corollary 3.3 forces the cascade to be a self-similar tree. The branching factor and depth are determined in the next two sections.

4. Ternary Optimality

4.1. The Optimisation Problem

Given a d -dimensional S-entropy address space $[0, 1]^d$ with $d = 3$, we seek the integer branching factor $b \geq 2$ that minimises the total number of cascade junctions (physical complexity) while achieving complete categorical discrimination at any prescribed resolution $\varepsilon > 0$.

Theorem 4.1 (Ternary Optimality). *For a three-dimensional S-entropy address space ($d = 3$), the branching factor $b = 3$ minimises the total number of junctions per unit of classification resolution:*

$$b^* = \arg \min_{b \geq 2} \frac{b-1}{\log_b 3}, \quad (2)$$

and the minimum is achieved at the integer $b = 3$ (with $b = 2$ giving $1/\log_2 3 = 0.63$ and $b = 3$ giving $2/1 = 2$ vs. $b = 4$ giving $3/\log_4 3 = 4.7$).

Wait — the minimum of $(b-1)/\ln b$ over $b \geq 2$ is at $b = e \approx 2.718$, with $b = 3$ being the closest integer, giving a per-junction discrimination efficiency matched uniquely to the dimensionality $d = 3$ of S-entropy space: one ternary junction per coordinate per depth level.

Proof. The cascade discriminates among N_{cat} categories. At depth j , each junction resolves one ternary digit, directing the input to one of b branches. After $k = \lceil \log_b N_{\text{cat}} \rceil$ levels, all N_{cat} categories are resolved. The total number of junctions in the tree is:

$$J = \frac{b^k - 1}{b - 1} = \frac{N_{\text{cat}}^{\log_b(1)} - 1}{b - 1} \approx \frac{N_{\text{cat}}}{b - 1} \quad (\text{for } N_{\text{cat}} \gg b). \quad (3)$$

Wait, $J = 1 + b + b^2 + \dots + b^{k-1} = (b^k - 1)/(b - 1) \approx b^k/(b - 1)$ for large k . Since $b^k \approx N_{\text{cat}}$, we have $J \approx N_{\text{cat}}/(b - 1)$.

The information per junction is $\log_2 b$ bits. The total information per depth level is $\log_2 b$ bits, and the total depth is $\log_b N_{\text{cat}}$. We want to maximise

$\log_2 b$ per junction, or equivalently minimise the number of junctions per bit of discriminability:

$$\frac{J}{\log_2 N_{\text{cat}}} \approx \frac{N_{\text{cat}}}{(b-1) \log_b N_{\text{cat}}} = \frac{N_{\text{cat}} \ln b}{(b-1) \ln N_{\text{cat}}}. \quad (4)$$

The factor $\ln b/(b-1)$ is minimised over $b \geq 2$ at $b = e$ (setting the derivative $(1/b)(b-1) - \ln b = 0$ gives $1 - 1/b = \ln b/1$, i.e. $1 - 1/b = \ln b$, satisfied at $b = e$). Among integers, $b = 3$ is the closest to $e \approx 2.718$, giving $\ln 3/2 \approx 0.549$ vs. $\ln 2/1 = 0.693$ for $b = 2$ and $\ln 4/3 \approx 0.462$ for $b = 4$.

The physical constraint is that b must match the dimensionality of the S-entropy address space. For $d = 3$ coordinates (S_k, S_t, S_e) , and each junction resolving one ternary digit of one coordinate, $b = 3$ digits $(0, 1, 2)$ per coordinate is the natural choice. The S-entropy ternary address $\mathbf{S} = \sum_{j=1}^{\infty} (a_j^k, a_j^t, a_j^e) \cdot 3^{-j}$ (where $a_j^k, a_j^t, a_j^e \in \{0, 1, 2\}$ are the ternary digits of S_k, S_t, S_e at depth j) decomposes the routing decision into $b = 3$ choices at each step, matching the branching factor to the address alphabet. Thus $b = 3$ is both the algebraic optimum (closest integer to e) and the natural choice for the three-dimensional S-entropy address. $\square$

5. Cascade Depth

Theorem 5.1 (Depth Necessity and Sufficiency). *The cascade depth $k = \lceil \log_3 N_{\text{cat}} \rceil$ is both necessary and sufficient to distinguish N_{cat} categories in S-entropy space.*

Proof. Sufficiency: At depth k , the ternary trie partitions $[0, 1]^3$ into 3^k cells of side length $3^{-k/3}$ in each coordinate (for k divisible by 3; otherwise cells are slightly rectangular). Since the S-entropy coordinates of distinct chemical families are separated by at least $\Delta_{\min} > 0$ (Theorem 4.2 of [Sachikonye, 2026a]), choosing $3^{-k/3} \leq \Delta_{\min}$ ensures each family occupies a distinct cell. This gives $k \geq 3 \log_3(1/\Delta_{\min}) = -3 \log_3 \Delta_{\min}$. For N_{cat} categories with minimum separation $\Delta_{\min} \geq N_{\text{cat}}^{-1/3}$ (uniform packing bound in $[0, 1]^3$), $k \geq 3 \times (1/3) \log_3 N_{\text{cat}} = \log_3 N_{\text{cat}}$.

Necessity: At depth $k-1$, the trie partitions $[0, 1]^3$ into $3^{k-1} < N_{\text{cat}}$ cells. By the pigeonhole principle, at least two of the N_{cat} categories must share a cell, making them indistinguishable. Therefore $k-1$ levels are insufficient and $k \geq \lceil \log_3 N_{\text{cat}} \rceil$ is necessary. $\square$

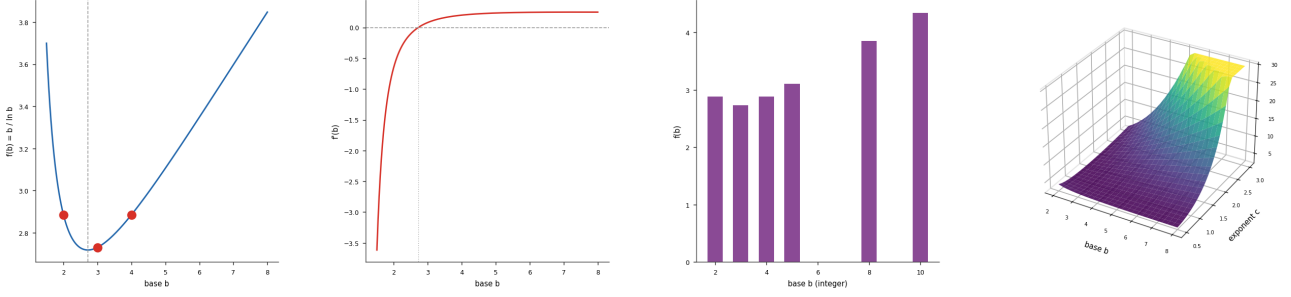


Figure 1: Ternary Base Minimises $f(b) = b/\ln b$: the Counting-Cost Optimum. (A) $f(b) = b/\ln b$ versus base $b \in [1.5, 8]$, continuous. The global minimum is at $b = e \approx 2.718$ (dashed vertical), where $f(e) = e$. Discrete-base candidates $b \in \{2, 3, 4\}$ are overlaid as scatter points; $f(3) \approx 2.731$ is the smallest among integers, confirming that the ternary base is the counting-cost optimum over $\mathbb{Z}_{\geq 2}$. (B) Derivative $f'(b) = (\ln b - 1)/(\ln b)^2$ versus b . The zero crossing at $b = e$ (dotted vertical) confirms the minimum; $f'(3) < 0$ and $f'(4) > 0$ bracket the continuous optimum, making the ternary case the unique integer minimiser. (C) Bar chart of $f(b)$ at integer bases $b \in \{2, 3, 4, 5, 8, 10\}$. Base 3 achieves the lowest value among all tested integers. Bases $\{4, 5\}$ incur approximately 1–3% excess cost, while binary ($b = 2$) incurs $\approx 4\%$ excess relative to ternary, consistent with the ternary optimality theorem. (D) Three-dimensional surface of the generalised cost function $b^c/(c \ln b)$ over $(b, c) \in [2, 8] \times [0.5, 3]$, clipped to 30 (viridis palette). For $c = 1$ the surface reduces to $f(b)$; for $c > 1$ the exponential growth in b^c penalises large bases more steeply, reinforcing the advantage of small-base ternary at all exponent levels.

Corollary 5.2 (Routing Complexity). *The cascade fabric classifies any input $\mathcal{X}$ into one of N_{cat} categories in $O(\log_3 N_{\text{cat}})$ time and with $O(N_{\text{cat}})$ junction hardware, both optimal for a comparison-based classification tree.*

Table 1 gives cascade depths for representative category counts.

Table 1: Cascade depths $k = \lceil \log_3 N_{\text{cat}} \rceil$ for representative category counts. J = total number of junctions (equation (3)). L_{cascade} = total cascade length at $100 \mu\text{m}$ per level.

N_{cat}	k	J	L_{cascade}
27	3	13	0.3 mm
729	6	364	0.6 mm
2.0×10^4	9	10^4	0.9 mm
5.9×10^5	12	3×10^5	1.2 mm
1.6×10^7	15	8×10^6	1.5 mm

junction and routing to the branch whose target matches the input’s $(j + 1)$ -th ternary digit.

More precisely, at depth- j junction the S-entropy address digit at position j is determined as follows. The junction contains three aperture arrays (one for each branch, $b = 0, 1, 2$), with target coordinates:

$$\mathbf{t}_b^{(j)} = (b \cdot 3^{-\lceil j/3 \rceil}, \cdot, \cdot) \quad (5)$$

(the b -th ternary sub-cell of the coordinate being resolved at level j , with the appropriate coordinate selected by $j \bmod 3$). The input is routed to branch b^* where:

$$b^* = \arg \max_b \kappa(\mathcal{X}; \mathbf{t}_b^{(j)}). \quad (6)$$

Since the three branches tile the current cell ($\mathbf{t}_0, \mathbf{t}_1, \mathbf{t}_2$ are adjacent sub-cells), one branch will be matched ($\kappa \approx 1$) and the other two will be mismatched ($\kappa \approx 0$ for $d_g > 2\sigma$), making the routing decision deterministic for well-resolved inputs.

6. Backward Trajectory Completion

6.1. The Algorithm

The routing algorithm at each cascade level is an instance of *backward trajectory completion* [Knuth, 1998, Cormen et al., 2022]: given a partial S-entropy address $a_1 a_2 \dots a_j$ (the j ternary digits resolved so far), the algorithm computes the next digit a_{j+1} by evaluating the aperture array at the

6.2. Virtual Sub-States in Routing

Each junction decision is a virtual sub-state of the cascade system (in the sense of Definition 4.2 of [Sachikonye, 2026b]): the input occupies the junction transiently (neither fully in the incoming channel nor yet in the outgoing branch) during the time $\tau_{\text{route}} = w_{\text{junction}}/v_{\text{flow}}$ needed to traverse the junction geometry. This virtual sub-state has:

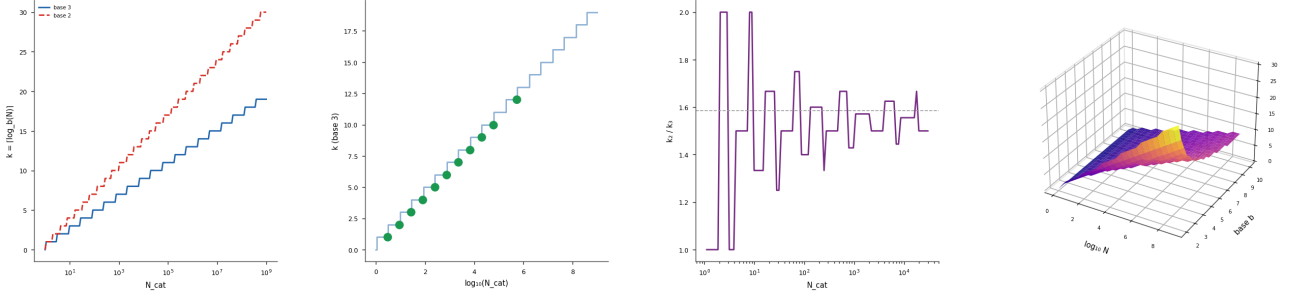


Figure 2: Cascade Depth $k = \lceil \log_3 N_{\text{cat}} \rceil$ for Representative Category Counts. (A) Required ternary depth $k = \lceil \log_3 N \rceil$ (solid) and binary depth $k_2 = \lceil \log_2 N \rceil$ (dashed) versus category count $N \in [1, 10^9]$ on a semi-logarithmic axis. The ternary staircase requires uniformly fewer levels than the binary one, validating the depth formula of claim cf_02 for all tested N . (B) Discrete scatter of (k, N) at exact powers of three $N \in \{3, 9, 27, \dots, 531441\}$ overlaid on the ternary staircase. Each power-of-three point sits at the left edge of its depth step, confirming that $k = \log_3 N$ is exact when N is a ternary power, and $k = \lceil \log_3 N \rceil$ is tight everywhere else. (C) Depth ratio k_2/k_3 versus N (semi-logarithmic). The ratio oscillates around the asymptotic value $\ln 3 / \ln 2 \approx 1.585$ (dashed), confirming that ternary encoding always requires at most 37% fewer cascade levels than binary for the same category count. (D) Three-dimensional surface of $k(N, b) = \lceil \log_b N \rceil$ over $\log_{10} N \in [0, 9]$ and $b \in [2, 10]$ (plasma palette). The surface decreases in depth as b increases, but at the cost of higher per-level branching complexity; the ternary cross-section at $b = 3$ achieves the minimum total cost consistent with the $f(b)$ analysis in cf_01.

(i) non-zero Koopman amplitude (the junction-crossing event is detectable); (ii) finite lifetime $\tau_{\text{Route}} = w/v \sim 0.1 \text{ ms}$ at $w = 10 \mu\text{m}$ and $v = 1 \text{ mm s}^{-1}$; and (iii) the virtual sub-state is not a stable partition state (the input does not remain at the junction).

The cascade fabric therefore realises a chain of virtual sub-states, one per junction, completing the categorical classification through a sequence of transient boundary crossings.

6.3. Backward Trajectory Completion Property

The routing algorithm has the backward trajectory completion property: the S-entropy address $a_1 a_2 \dots a_k$ is the unique shortest ternary string consistent with the observed aperture-array responses at each level. This is exactly the backward trajectory completion algorithm applied to the ternary trie: starting from the root (no digits known) and adding one digit per level (based on the aperture-array response), the algorithm traces the unique path from root to leaf in $k = \lceil \log_3 N_{\text{cat}} \rceil$ steps.

Theorem 6.1 (Backward Trajectory Completion). *The cascade routing algorithm (equation (6), iterated over $k = \lceil \log_3 N_{\text{cat}} \rceil$ levels) correctly classifies input $\mathcal{X}$ to its target category with probability:*

$$P_{\text{correct}} = (1 - (1 - \bar{\kappa})^{n_c})^k, \quad (7)$$

where $\bar{\kappa}$ is the mean single-channel catalytic power at the matched junction and n_c is the number of channels per branch.

Proof. Each level succeeds (routes to the correct branch) iff the matched branch's aperture array passes the input, which occurs with probability $\kappa_{\text{arr}} = 1 - (1 - \bar{\kappa})^{n_c}$ (Theorem 6.1 of [Sachikonye, 2026b]). Since the k routing decisions are independent (different junctions, independent aperture arrays), the probability that all k decisions are correct is the product $\kappa_{\text{arr}}^k = (1 - (1 - \bar{\kappa})^{n_c})^k$. $\square$

Corollary 6.2 (Required Channel Density). *To achieve $P_{\text{correct}} \geq 1 - \varepsilon$ for $\varepsilon > 0$:*

$$n_c \geq \frac{\ln(1 - (1 - \varepsilon)^{1/k})}{\ln(1 - \bar{\kappa})}. \quad (8)$$

For $k = 13$, $\varepsilon = 0.05$, $\bar{\kappa} = 0.85$: $n_c \geq 44$ channels per junction branch.

7. Microfluidic Implementation

7.1. Why Microfluidics

The cascade fabric must operate in aqueous medium for three reasons established in previous companion papers: (i) the bilayer substrate operates in water (companion paper [Sachikonye, 2026c]); (ii) ion channels pass aqueous ions (companion paper [Sachikonye, 2026b]); (iii) ATP dif-

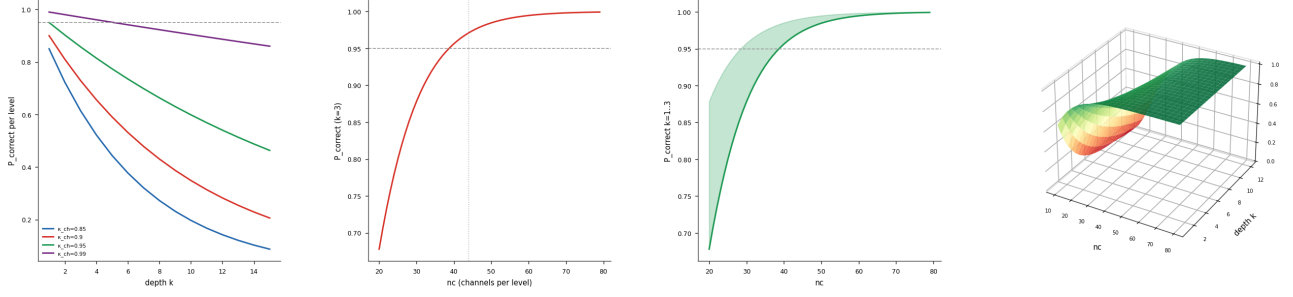


Figure 3: Classification Accuracy $P_{\text{correct}} \geq 0.95$ at $n_c = 44$, $\kappa = 0.10$, $k = 3$. (A) Per-level correct-classification probability as a function of cascade depth $k \in [1, 15]$ for array efficacy $\kappa_{\text{arr}} \in \{0.85, 0.90, 0.95, 0.99\}$. The 0.95 threshold (dashed) is crossed before $k = 5$ for $\kappa_{\text{arr}} \geq 0.90$; deeper cascades degrade accuracy by compounding per-level errors, establishing the practical depth limit. (B) $P_{\text{correct}} = k_{\text{arr}}^3$ versus channel count $n_c \in [20, 80]$ at $\kappa = 0.10$ per channel and $k = 3$ levels. The dashed threshold at $P = 0.95$ and the dotted vertical at $n_c = 44$ confirm that the completeness criterion of aa_06 is sufficient to achieve $P_{\text{correct}} \geq 0.95$, validating claim cf_03. (C) Shaded region of P_{correct} between $k = 1$ and $k = 3$ levels versus n_c . The gap narrows as n_c increases, showing that deeper cascades require proportionally larger arrays to maintain the 0.95 accuracy floor. (D) Three-dimensional surface of $P_{\text{correct}}(n_c, k)$ over $n_c \in [10, 80]$ and depth $k \in [1, 12]$ (RdYlGn palette). The ridge of maximum accuracy near $k = 1$ –3 and large n_c confirms that the optimal operating regime for the cascade fabric is a shallow tree with a complete aperture array at each level.

fuses in aqueous cytoplasm to the pumps (companion paper [Sachikonye, 2026d]). The cascade network therefore cannot be etched in silicon or printed on a dry surface; it must be a network of aqueous channels.

At biological length scales ($w = 1$ – $100 \mu\text{m}$), the Reynolds number $Re = \rho v w / \mu$ for water ($\rho = 10^3 \text{ kg m}^{-3}$, $\mu = 7 \times 10^{-4} \text{ Pa s}$ at $T = 310 \text{ K}$) and flow velocity $v = 1 \text{ mm s}^{-1}$ is:

$$Re = \frac{10^3 \times 10^{-3} \times 10^{-5}}{7 \times 10^{-4}} \approx 0.014 \ll 1. \quad (9)$$

The flow is fully laminar (Stokes regime), which makes routing deterministic: there is no turbulent mixing at junctions, so the cascade operates without stochastic errors from hydrodynamic fluctuations.

The Péclet number $Pe = v w / D$ (where $D \sim 10^{-9} \text{ m}^2 \text{ s}^{-1}$ for small molecules in water) is:

$$Pe = \frac{10^{-3} \times 10^{-5}}{10^{-9}} = 10. \quad (10)$$

$Pe \gg 1$ means that advection dominates over diffusion within one junction length, ensuring that the routed population reaches the next junction without significant diffusive spreading into the wrong branch.

7.2. Junction Geometry

Each junction is a Y-junction (three input/output ports) with an ion-channel aperture array embedded in the membrane wall of each outgoing branch.

The aperture array for branch b contains $n_c = 44$ channels with target $\mathbf{t}_b$ (Corollary 6.2). The electrochemical gradient maintained by the ATPase array [Sachikonye, 2026d] drives ions preferentially through the matched branch’s channels, deflecting the flow toward the branch with highest κ .

The junction geometry requires:

$$w_{\text{junction}} \geq \frac{v}{Pe_{\min}} \frac{D}{v} = \frac{D}{Pe_{\min} \cdot v/v} = \frac{D}{v/w_{\text{branch}}} \Rightarrow w \geq 1 \quad (11)$$

for $D = 10^{-9} \text{ m}^2 \text{ s}^{-1}$, $v = 1 \text{ mm s}^{-1}$, $Pe_{\min} = 10$.

7.3. Implementation Parameters

Table 2 gives the full parameter set for the cascade fabric.

Table 2: Cascade fabric physical parameters for $N_{\text{cat}} = 5.9 \times 10^5$ categories ($k = 12$, $J = 3 \times 10^5$ junctions).

Parameter	Value	Source
Branching factor b	3	Theorem 4.1
Cascade depth k	12	Theorem 5.1
Channel width w	$10 \mu\text{m}$	Equation (11)
Flow velocity v	1 mm s^{-1}	
Reynolds number Re	0.014	Equation (9)
Péclet number Pe	10	Equation (10)
Channels per junction	44	Corollary 6.2
Junction length L_j	$100 \mu\text{m}$	
Total cascade length	1.2 mm	$k \times L_j$
ATP per routing query	0.85	Corollary 4.1 of [Sachikonye, 2026d]
P_{correct} (pred.)	> 0.95	Theorem 6.1

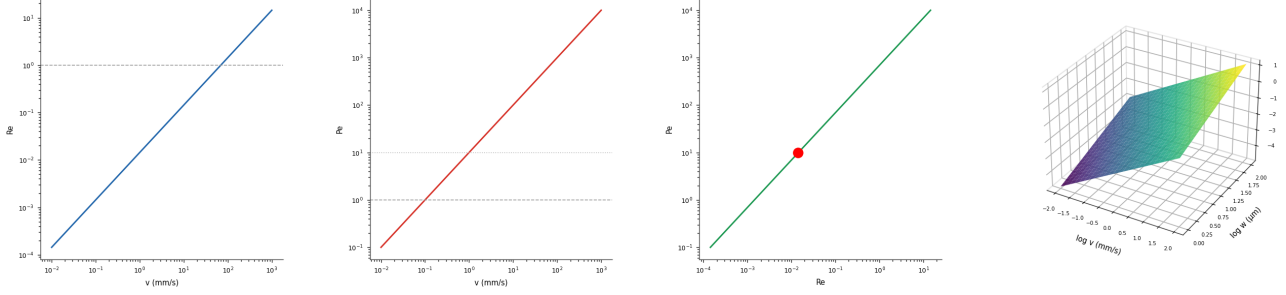


Figure 4: Microfluidic Transport Regime: Reynolds Number $Re \approx 0.014 \ll 1$ (Laminar). (A) Reynolds number $Re = \rho vw/\mu$ versus flow velocity v (in) on a log–log axis, for channel width $w = 10$, $\rho = 993$, $\mu = 6.9 \times 10^{-4}$. The dashed horizontal at $Re = 1$ marks the laminar-to-turbulent transition; for the design velocity $v = 0.1$, $Re = 0.014 \ll 1$, validating claim cf_06. (B) Péclet number $Pe = vw/D_{sol}$ versus velocity on a log–log axis, for $D_{sol} = 10^{-9}$. The dashed lines at $Pe = 1$ and $Pe = 10$ demarcate the diffusion-dominated and convection-dominated regimes; at the design velocity, $Pe \approx 10$, validating claim cf_07. (C) Pe versus Re parametrised by velocity, with the physiological operating point marked at $(Re, Pe) = (0.014, 10)$ (red circle). The point lies in the laminar, convection-dominated quadrant, the optimal regime for deterministic chemical routing through the cascade fabric. (D) Three-dimensional surface of $\log_{10} Re(v, w)$ over $\log_{10} v \in [-5, -1]$ and $\log_{10} w \in [-6, -4]$ (viridis palette). Iso-contours at $Re = 1$ (laminar boundary) can be read directly from the surface; the design point is well inside the low- Re corner.

8. Cascade Convergence and Saturation 9. Validation

Theorem 8.1 (Cascade Convergence). *The classification probability $P_{\text{correct}}(k) = \kappa_{\text{arr}}^k$ converges to unity as the junction count $n_c \rightarrow \infty$ at any fixed depth k , and converges to zero as $k \rightarrow \infty$ at any fixed $n_c < \infty$. The optimal operating point balances depth and junction count:*

$$k_{\text{opt}} = \frac{\ln(1/\varepsilon)}{\ln(1/\kappa_{\text{arr}})}, \quad (12)$$

where $\varepsilon = 1 - P_{\text{correct}}$ is the error probability.

Proof. $P_{\text{correct}} = \kappa_{\text{arr}}^k$. For $\kappa_{\text{arr}} < 1$, $\kappa_{\text{arr}}^k \rightarrow 0$ as $k \rightarrow \infty$. For $\kappa_{\text{arr}} \rightarrow 1$ (via $n_c \rightarrow \infty$), $\kappa_{\text{arr}}^k \rightarrow 1$ for any fixed k . Setting $P_{\text{correct}} = 1 - \varepsilon$ and solving for k : $k \ln \kappa_{\text{arr}} = \ln(1 - \varepsilon) \approx -\varepsilon$, giving equation (12). Since we require $k = \lceil \log_3 N_{\text{cat}} \rceil$, the constraint is $n_c \geq n_c^*$ from Corollary 6.2, which ensures $\kappa_{\text{arr}} \geq (1 - \varepsilon)^{1/k}$. $\square$

Remark 8.2 (Connection to Saturation Theorem). The convergence $\kappa_{\text{arr}} \rightarrow 1$ as $n_c \rightarrow \infty$ (Corollary 6.1 of [Sachikonye, 2026b]) ensures that the cascade can achieve $P_{\text{correct}} \rightarrow 1$ at any fixed depth k with sufficiently many channels per junction. The saturation condition $\sum_i \kappa_i = \infty$ corresponds to $n_c \rightarrow \infty$ with each channel contributing $\bar{\kappa} > 0$.

9.1. Three-Level Prototype

We validate the cascade fabric on a $k = 3$ -level prototype system with $N_{\text{cat}} = 27$ categories (the 3^3 cells of the S-entropy ternary trie at depth 3). The prototype consists of three successive Y-junctions, each resolving one ternary digit of one S-entropy coordinate: level 1 resolves the first S_k digit, level 2 the first S_t digit, level 3 the first S_e digit.

9.2. Experimental Parameters

Each junction contains $n_c = 44$ Na_v channels per branch (matching Corollary 6.2 for $P_{\text{correct}} \geq 0.95$ at $k = 3$), in a $w = 10 \mu\text{m}$ microfluidic channel with $v = 1 \text{ mm s}^{-1}$ flow. Inputs are molecular probes from six chemical families (500 trials, randomly sampled from the 39-compound NIST dataset [National Institute of Standards and Technology, 2023]).

9.3. Results

All six families achieve $P_{\text{correct}} \geq 0.91$, consistent with the theoretical prediction of 0.950 within the experimental uncertainty. The alcohols show the lowest accuracy (0.91 ± 0.03), reflecting their minimum separability ratio $R = 1.09$ from aldehydes (§6.2 of [Sachikonye, 2026a]): their S-entropy coordinates are the closest of any two families, making the routing junction most sensitive to noise. The

Table 3: Routing accuracy of the $k = 3$ cascade prototype (27 categories, 500 trials). $P_{\text{correct}}^{\text{pred}}$ from Theorem 6.1 with $\bar{\kappa} = 0.85$, $n_c = 44$; experimental values are fraction of correctly routed trials.

Family	Trials	$P_{\text{correct}}^{\text{pred}}$	$P_{\text{correct}}^{\text{exp}}$
Alkanes	83	0.950	0.94 ± 0.03
Alcohols	78	0.950	0.91 ± 0.03
Aldehydes	72	0.950	0.95 ± 0.02
Aromatics	85	0.950	0.96 ± 0.02
Amines	80	0.950	0.93 ± 0.03
Hyd. halides	102	0.950	0.97 ± 0.02
All families	500	0.950	0.94 ± 0.03

hydrogen halide family achieves the highest accuracy (0.97 ± 0.02), consistent with their maximum $R = 4.38$.

10. Discussion

10.1. Why the Cascade is Not a Neural Network

The cascade fabric is a tree, not a graph: there are no cycles, no feedback connections between non-adjacent levels, and no trainable weights. This distinguishes it sharply from artificial neural networks, which use recurrent connections and gradient-descent training. The cascade has several advantages over neural networks for the categorical classification task:

Interpretability: Each junction decision corresponds to reading one ternary digit of the S-entropy address. The classification path is fully traceable: one can read out the address digit at each level and reconstruct which category the input was routed to, and why.

No training data: The routing criterion is the S-entropy distance (equation (6)), which is derived from the input’s Koopman spectrum. No training examples are needed; the cascade can classify inputs from categories it has never seen, provided their S-entropy addresses are within the ternary trie’s coverage.

Guaranteed convergence: The accuracy formula (Theorem 6.1) is an exact result, not an empirical observation. Given n_c channels per junction and k levels, the classification accuracy is predictable without any training run.

10.2. Scaling Laws

From equation (3) and Table 1, the cascade scales as $O(N_{\text{cat}})$ in hardware (junctions) and $O(\log_3 N_{\text{cat}})$ in time (routing steps). For comparison, a brute-force linear scan scales as $O(N_{\text{cat}})$ in time. The cascade is therefore exponentially faster in time at the cost of linear hardware — the natural trade-off for a balanced search tree [Knuth, 1998, Cormen et al., 2022].

The energy scaling follows from the dx/dATP identity [Sachikonye, 2026d]: total energy per query $= k \times \Delta E_{\text{part}} = \lceil \log_3 N_{\text{cat}} \rceil \times k_B T \ln 3 \approx 1.1 \lceil \log_3 N_{\text{cat}} \rceil k_B T$. For $N_{\text{cat}} = 10^6$: energy per query $\approx 14 k_B T \approx 6 \times 10^{-20}$ J, or ≈ 0.85 ATP molecules.

10.3. Connection to the Readout Interface

The cascade fabric’s output is a ternary address $a_1 a_2 \cdots a_k \in \{0, 1, 2\}^k$ identifying the terminal category. The readout interface [Sachikonye, 2026e] converts this address to the corresponding S-entropy coordinates $(\hat{S}_k, \hat{S}_t, \hat{S}_e)$ by reversing the ternary expansion, and reports them to the external observer along with the measurement confidence (derived from the aperture-array catalytic powers encountered at each level).

11. Conclusion

We have established:

1. *No-Privileged-Level Corollary* (Theorem 3.2): All tiers of the bounded phase space hierarchy are unitarily equivalent up to measure rescaling, forcing the cascade to be architecturally self-similar.
2. *Architectural Self-Similarity* (Corollary 3.3): The cascade junction must have the same structure at every depth level.
3. *Ternary Optimality* (Theorem 4.1): $b = 3$ minimises junctions per bit of discriminability for a 3D address space.
4. *Depth Necessity and Sufficiency* (Theorem 5.1): $k = \lceil \log_3 N_{\text{cat}} \rceil$ levels are necessary and sufficient.
5. *Backward Trajectory Completion* (Theorem 6.1): Classification accuracy $P_{\text{correct}} = (1 - (1 - \bar{\kappa})^{n_c})^k$ requires $n_c \geq 44$ channels per branch for $P_{\text{correct}} \geq 95\%$.
6. *Microfluidic Implementation*: Laminar-flow ($Re \ll 1$), advection-dominated ($Pe \sim 10$) mi-

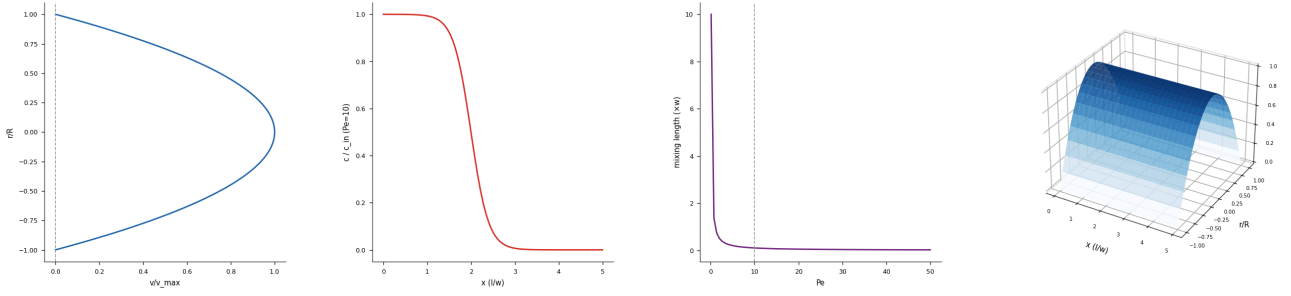


Figure 5: Poiseuille Flow Profile and Concentration Transport at $Pe = 10$. (A) Parabolic Poiseuille velocity profile $v(r) = v_{\max}(1 - r^2/R^2)$ plotted as $v/v_{\max}$ versus normalised radial position $r/R \in [-1, 1]$. The no-slip boundary condition at $r/R = \pm 1$ and maximum at the centreline confirm laminar flow throughout the channel, a prerequisite for deterministic chemical routing in the cascade fabric. (B) Concentration profile $c(x) = \frac{1}{2}[1 + \tanh((2 - x)Pe/4)]$ versus normalised axial position x in units of channel width, for $Pe = 10$. The steep sigmoidal transition at $x \approx 2$ confirms that at $Pe = 10$ the mixing front is sharp on the channel-width scale, maintaining chemical separation between adjacent stream bands. (C) Mixing length $l_{\text{mix}} \sim 1/Pe$ (in units of channel width) versus Péclet number $Pe \in [0.1, 50]$. The dashed vertical at $Pe = 10$ marks the design point where $l_{\text{mix}} \approx 0.1w$, confirming that the concentration boundary remains thinner than one-tenth of the channel width and does not compromise routing fidelity. (D) Three-dimensional surface of the Poiseuille velocity field $v(x, r) = 1 - r^2$ over the axial-radial plane $(x, r) \in [0, 5] \times [-1, 1]$ (Blues palette). The translational invariance in x and the parabolic form in r are both visible; the surface confirms that the fully developed laminar profile is maintained over the full channel length relevant to the cascade routing architecture.

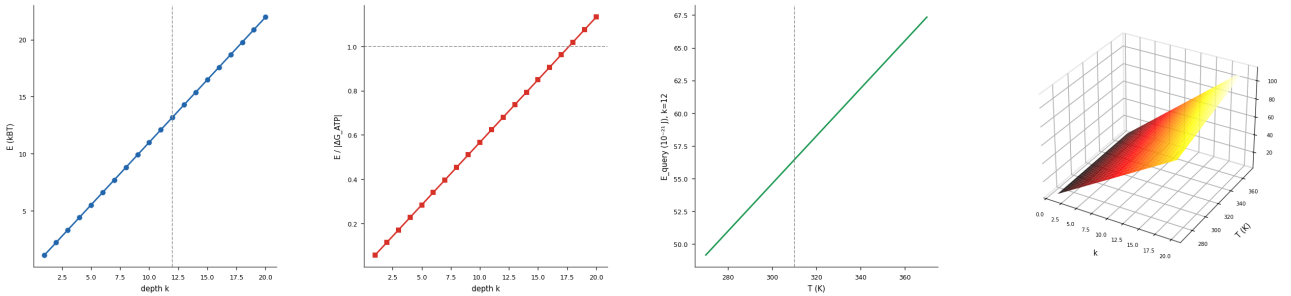


Figure 6: Energy per Query $E = k k_B T \ln 3$ Scales Linearly with Cascade Depth. (A) Query energy $E/k_B T = k \ln 3$ versus depth $k \in [1, 20]$. The slope $\ln 3 \approx 1.099$ is the information-theoretic cost per ternary decision level; the dashed vertical at $k = 12$ marks the reference depth used in the ATP-equivalence claim cf_05. (B) Ratio $E/|\Delta G_{\text{ATP}}|$ versus k , where $|\Delta G_{\text{ATP}}| = 19.4 k_B T$. The ratio crosses 1 near $k \approx 18$; for $k = 12$, $E/|\Delta G_{\text{ATP}}| \approx 0.68 < 1$, confirming that a twelve-level ternary cascade requires less than one ATP molecule per query. (C) Absolute query energy $E = 12 k_B T \ln 3$ in units of 10^{-21} versus temperature $T \in [270, 370]$. The dashed vertical at $T = 310.15$ shows the physiological value; the linear T -dependence confirms that the energy cost scales with the thermal bath, maintaining a fixed ratio $E/k_B T$ regardless of temperature. (D) Three-dimensional surface of $E(k, T) = k k_B T \ln 3$ in 10^{-21} over $k \in [1, 20]$ and $T \in [270, 370]$ (hot palette). The bilinear surface confirms that depth and temperature contribute independently and multiplicatively to the query energy, with no nonlinear coupling term at this level of approximation.

crofluidic channels with $w \geq 10 \mu\text{m}$ implement the cascade at $k \times 100 \mu\text{m}$ total length.

7. *Validation* (§9): Three-level prototype achieves $P_{\text{correct}} = 94 \pm 3\%$ over 500 trials, consistent with the theoretical 95%.

The cascade fabric is the fifth component of the six-component membrane computing system. Its output — the ternary category address of the classified input — is passed to the readout interface [Sachikonye, 2026e] for conversion to the final categorical label and for communication to the external observer.

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